Module F

Introduction to Machine Drawing

Chapter 19 Multiview Projection of Objects

Chapter 20 Auxiliary View of Objects

Chapter 21 Visualization of Objects

Chapter 22 Sectional Views of Objects

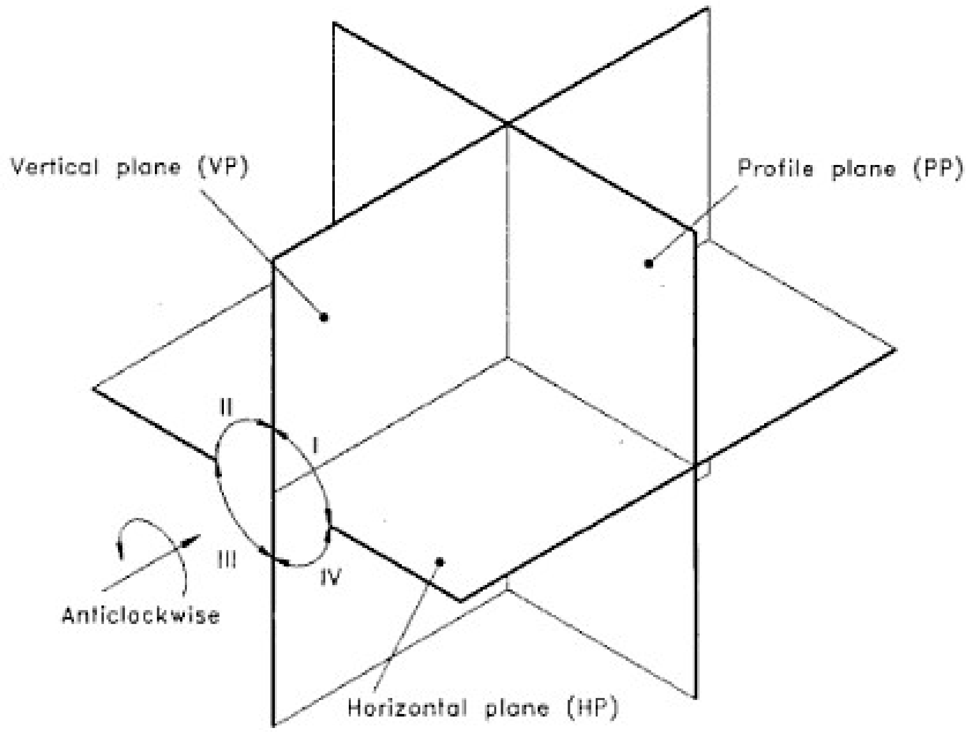


Figure 19.1 Planes of projection showing octants (anticlockwise system).

19.2 FIRST ANGLE PROJECTION AND THE LOCATIONS OF THE THREE VIEWS

In first angle projection, the object is assumed to be positioned in the first quadrant as shown in Figure 19.2. Here, the object is placed in such a way that its main faces are parallel to the principal planes and hence the projections of these faces on the principal planes will have the true shape and size.

The three views formed on the principal planes are described below:

1. Front view: The view of the object formed on the vertical plane (VP) or frontal plane, when looked orthogonally at the object in the direction marked FRONT, is called *front view*. Here, the VP is behind the object. Thus, the object is in between the plane of projection and the eye. This is indicated by

EYE > OBJECT > PLANE

2. Top view: The view of the object formed on the horizontal plane (HP) when looked orthogonally at the object from the top in the direction marked TOP, is called *top view*. Here, the horizontal plane is below the object. Thus, the object is in between the plane of projection and the eye. This is indicated by

EYE > OBJECT > PLANE

3. Left side view: The view of the object formed on the profile plane (PP), when looked orthogonally at the object in the direction marked LEFT-HAND SIDE, is called *left side view*. Here, the profile plane is behind the object. Thus, the object is in between the eye and the plane of projection. This is also indicated by

EYE > OBJECT > PLANE

To bring the three views into a single plane, revolve the coordinate planes through 90°, as indicated by the arrows. The complete layout of the three views of the object, after rabation, will be as shown in Figure 19.3.

19.3 THIRD ANGLE PROJECTION AND THE LOCATIONS OF THE THREE VIEWS

In third angle projection, the object is assumed to be positioned in the third quadrant. Here, the object is placed in such a way that its main faces are parallel to the principal planes and hence the projections of these faces on the principal planes will have the true shape and size.

The complete layout of the three views of an object in third angle projection is as shown in Figure 19.4.

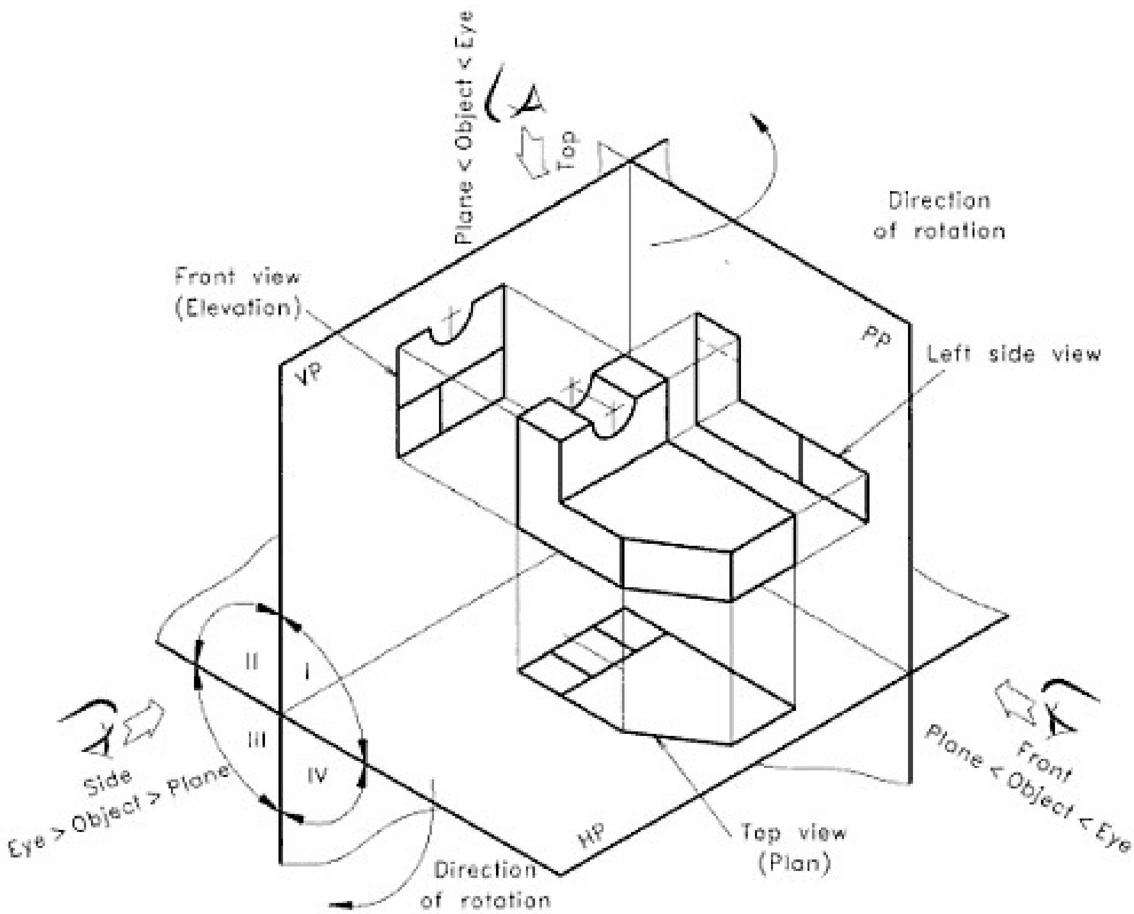


Figure 19.2 First angle projection following anticlockwise system.

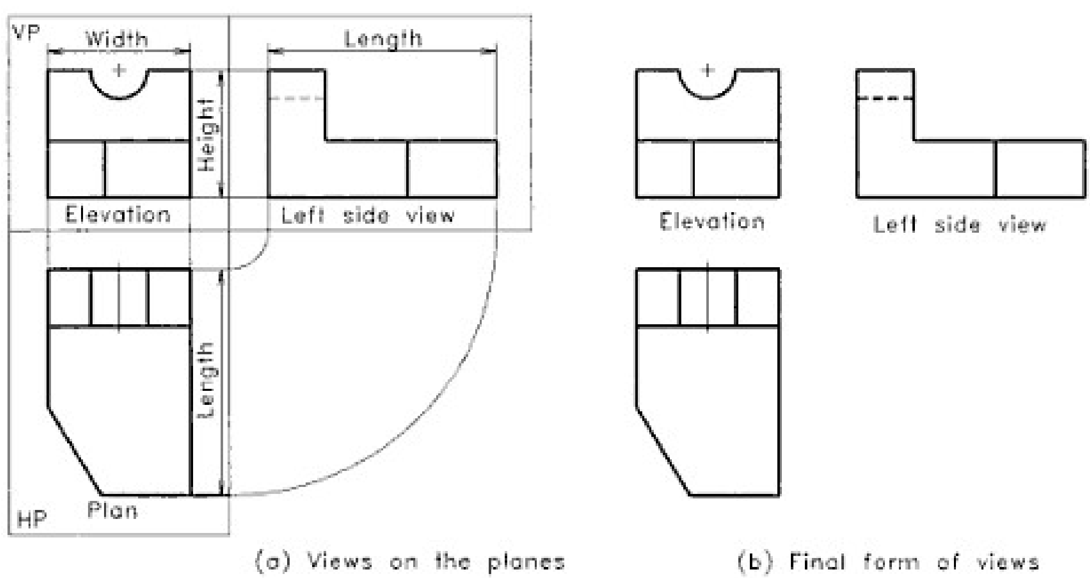


Figure 19.3 Layout of the principal views (first angle projection).

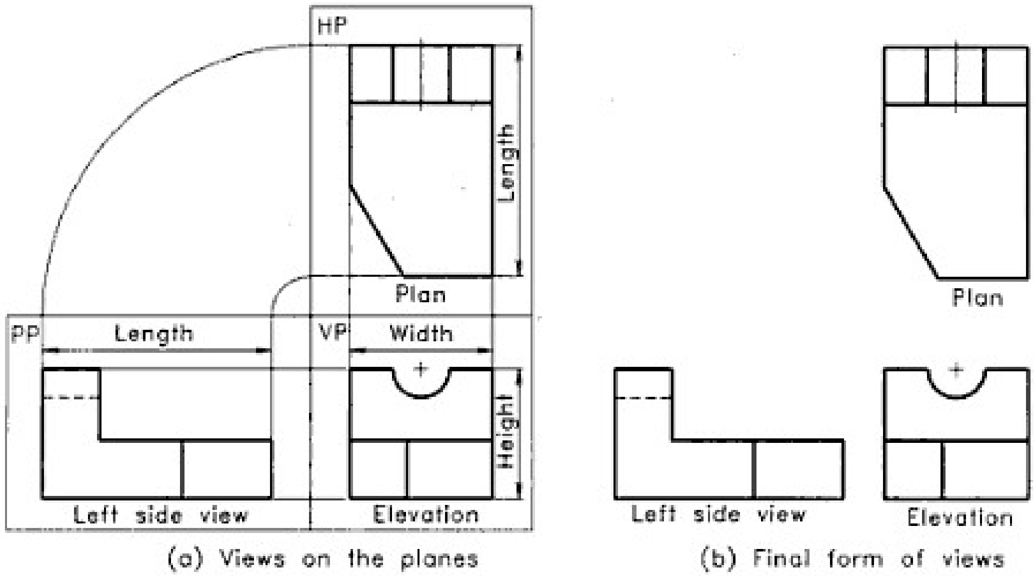


Figure 19.4 Layout of the principal views (third angle projection).

19.4 TRANSPARENT BOX AND THE SIX ORTHOGRAPHIC VIEWS

To describe the shape and size of a complicated object completely on a sheet of paper, sometimes, more than three views are required. In such cases, transparent box method can be used to get six different views of the object. Here, the object is assumed to be placed inside a transparent box, keeping its important face parallel to the front side of the box (see Figure 19.5). The six sides of the box are assumed to be six planes of projection. The observer views the enclosed object from outside. Six views are obtained on the six planes by drawing projectors from various points on the object to these planes. These views are called *front, top , right side, left side, bottom* and *rear views*. To transfer these six views, the box is opened to one plane, the plane of the drawing sheet. The six views can be developed by applying the principle of first and third angle projection methods.

In first angle projection method, the object is placed between the eye and the plane of projection. Hence, we follow the *EYE > OBJECT > PLANE* principle.

Consider a transparent box ABCDEFGH containing an object inside it, as shown in Figure 19.5. The following are the views obtained on the six sides of the transparent box.

1. *Front view:* The view of the object formed on the rear side ABCD of the box, when looked in the direction of the arrow marked by FRONT, is called *front view*.

2. *Top view:* The view of the object formed on the bottom side DCGH of the box, when looked in the direction of the arrow marked by TOP, is called *top view*.

3. *Left side view:* The view of the object formed on the right side BFGC of the box when looked in the direction of the arrow marked by LEFT SIDE, is called *left side view*.

4. *Right side view:* The view of the object formed on the left side AEHD of the box, when looked in the direction of the arrow marked by RIGHT SIDE, is called *right side view*.

5. *Bottom view:* The view of the object formed on the top side ABFE of the box, when looked in the direction of the arrow marked by BOTTOM, is called *bottom view*.

6. *Rear view:* The view of the object formed on the front side EFGH of the box, when looked in the direction of the arrow marked by REAR, is called *rear view*.

Assume that the transparent box is formed by hinging the sides of the box onto the edges of these sides. It may be noted that all the sides of the transparent box except the front side EFGH, are hinged to the four edges AB, BC, CD and DA of the rear side ABCD. The front side EFGH is hinged to the edge FC. There are two hinges on each side. Now to open the box, rotate the sides of the box outwards about the respective hinges as shown in Figure 19.6. All the sides of the box are opened out in such a way that the front view occupies the central position. Continue the process of the rotation until all sides of the box lie in a single plane, the plane of the drawing sheet.

The layout of the six views in first angle projection method is shown in Figure 19.7. The rear view may also be placed to the left-hand side of the right side view.

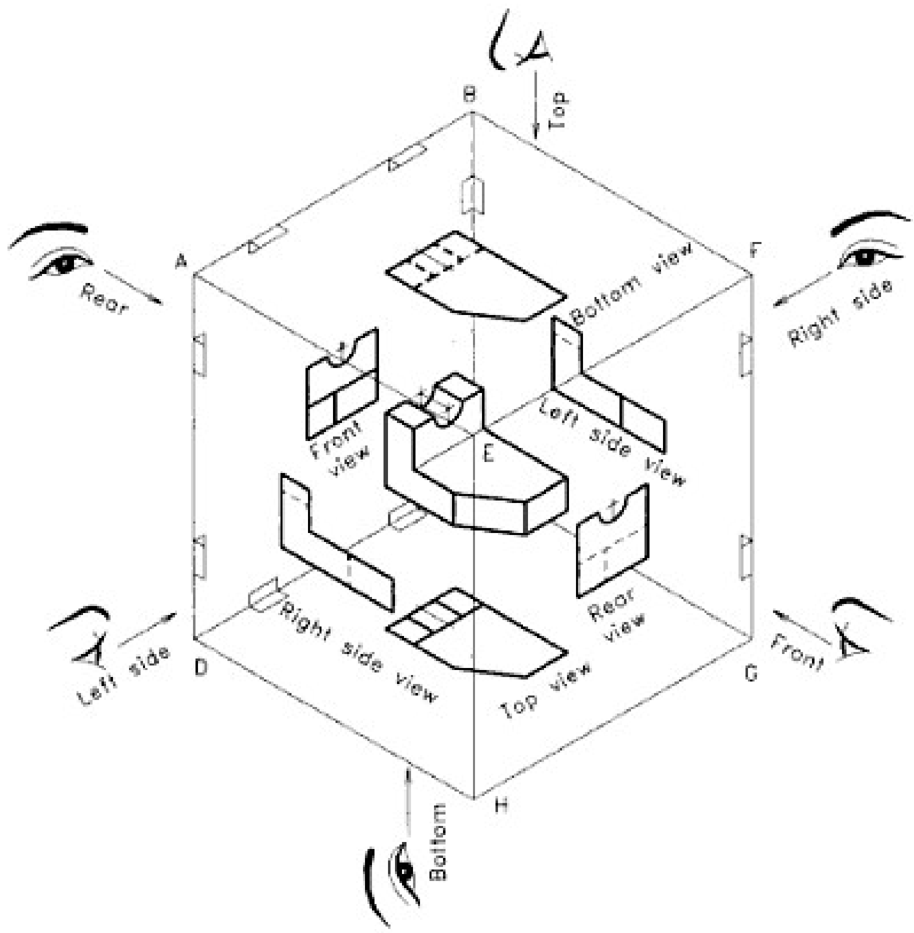


Figure 19.5 Transparent box containing an object (first angle projection).

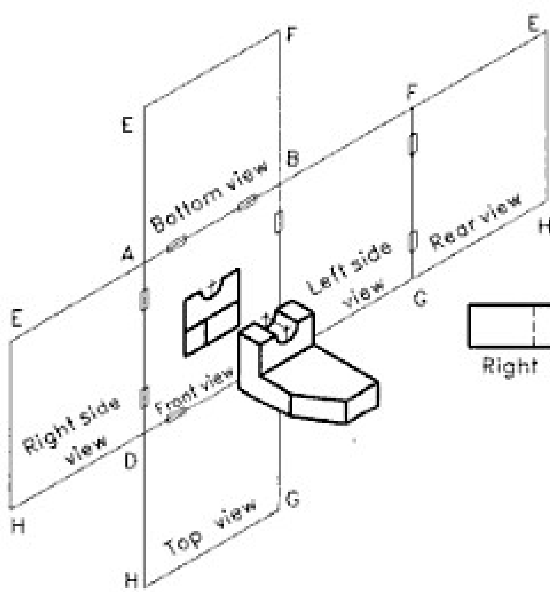


Figure 19.6 Opening of the transparent box (first angle projection).

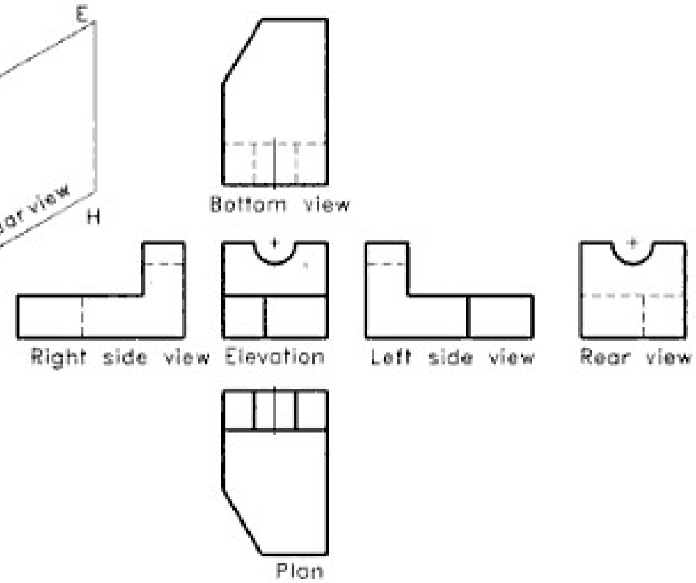


Figure 19.7 Layout of the six views (first angle projection).

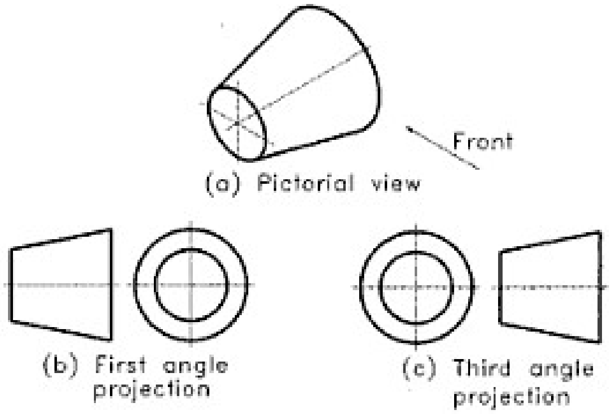


Figure 19.8 Symbol for first angle poojection.

19.5 INDICATION OF FIRST ANGLE PROJECTION

The method of projection used, must be indicated inside the space provided in the title block of the drawing sheet. A distinguishing symbol is recommended by the Bureau of Indian Standards for this purpose. The front and left side views of a frustum of a cone lying with its axis horizontal, is used for this. The symbol for first angle projection is shown in Figure 19.8.

19.6 SELECTION OF MINIMUM NUMBER OF VIEWS

For describing the shape of an object completely by its orthographic views, it is necessary to select the number of views required, and combine.

The number of views required for describing an object clearly and completely depends upon the extent of complexity involved in it. Based upon the number of views required, the drawings can be classified into the following categories:

1. One-view drawing
2. Two-view drawing
3. Three-view drawing

It may be noted that only minimum number of views, that will describe the object clearly and completely, should be drawn.

One-view Drawing

An object having cylindrical, square or hexagonal features can be completely described by a single orthographic view. Such a drawing is called *one-view drawing*. Here, the features are expressed by a note or an abbreviation.

In Figure 19.9, the cylindrical part is indicated by the notation ϕ and the square part is indicated by the notation □. The square part is identified by drawing thin crossed diagonal lines on the feature. Plate of any size can be described by a single orthographic view. The thickness of the plate may be expressed by a note.

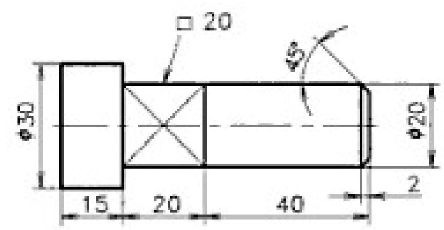


Figure 19.9 A pin (one view drawing).

Two-view Drawing

Objects which are symmetrical about two axes can be represented clearly and completely by two views. Such a drawing is called *two-view drawing*.

The largest face, showing most of the details and having minimum number of hidden lines, is selected as the front view. The second view may be the top or the side view.

It may be noted that any two views will not be sufficient to describe an object completely. Proper combination of the views should be selected. Isometric views of three prisms and the plan of these prisms are shown in Figure 19.10(a). The front and top views of these prisms shown in Figure 19.10(b) are not sufficient to describe them completely. But, the front and side views of these prisms describe the objects clearly and completely. Side views of the prisms are shown in Figure 19.10(c).

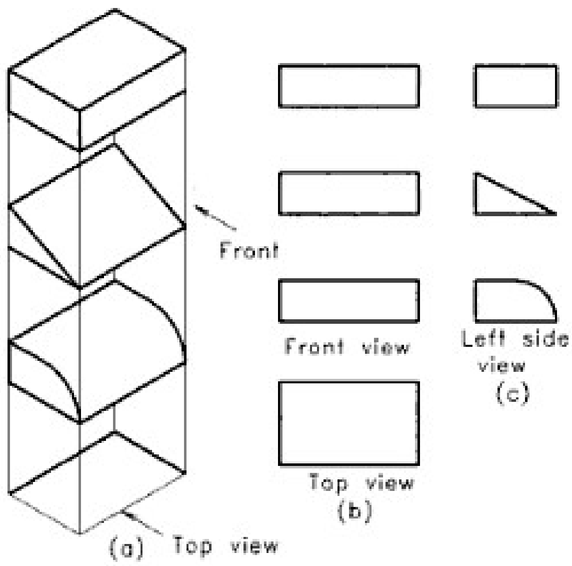


Figure 19.10 Two-view drawing of prisms.

Three-view Drawing

Most of the objects can be represented clearly and completely by three views. Such a drawing is called *three-view drawing*. The largest face showing most of the details is selected as the front view. Here, the object is placed in its functional position as far as possible and with the principal faces parallel to the planes of projection. A three view drawing of a cast iron block is shown in Figure 19.11.

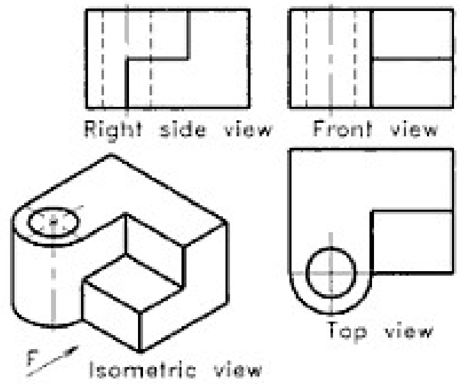


Figure 19.11 Three view drawing of a cast iron block.

19.7 USE OF LINE TYPES AND DIMENSIONING OF VIEWS

Orthographic views are drawn with proper line types as allowed by BIS. Figure 19.12 shows three orthographic views of an object drawn in first angle projection with proper line types and dimensioning. The pictorial view is also given for reference. The important points to be considered are given below.

Line Types for Orthographic View

1. The visible edges and all the outermost edges and surfaces are represented by continuous thick lines, i.e. Type A lines.

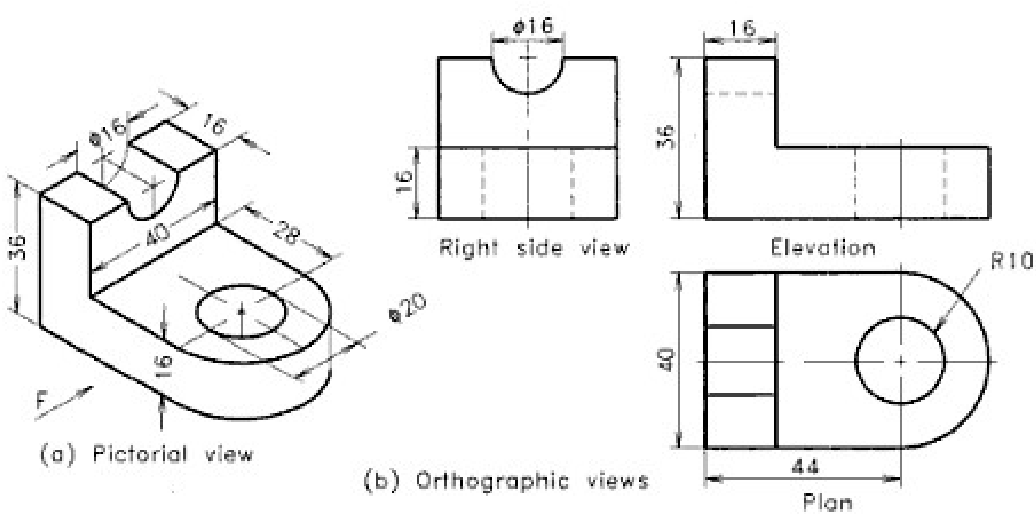


Figure 19.12 A machine part.

2. The hidden details are shown only if they are required. They are represented by short dashes using Type E or F lines. The method of drawing hidden lines and the rules for superimposing them are explained in Chapter 1.
3. Thin chain line (Type G) is used to represent centre lines and lines of symmetry.
4. Orthographic projections of machine parts generally have holes, circles, lines of symmetry, etc. Hence, the fixing of the view location and the further construction of shapes are progressed only after drawing all the important centre lines of the related views.
5. As a rule, a circular hole or projection should be drawn with the centre lines in horizontal and vertical directions. On a pitch circle, the hole centre is represented by the pitch circle drawn with chain line, and an intersecting radial chain line is drawn from the centre of the pitch circle.
6. Thin continuous line (Type B) is used for sketching views, section lines, construction lines and dimension lines.
7. Projection lines and reference lines are assumed to be invisible in the orthographic views of objects, even though they are compulsory for orthographic views of solids. It has to be noted that projection lines are not drawn, but the views and their details should lie in the exact alignment obeying the rules of projection.

Dimensioning of Orthographic Views

1. Dimensioning of orthographic views of objects is done by following Method–1, as described in Chapter 4. Method-2 is also permitted by BIS. Since Method-1 has certain advantages, it is generally followed for machine drawing.
2. The dimension lines are drawn using thin continuous (Type B) lines and the text is printed using thick single stroke letters as explained in Chapter 2.
3. The complete dimensional values have to be shown on the related orthographic views. They may be distributed in all the related views almost evenly.
4. Writing dimensional values on hidden details and over the view should be avoided.
5. There is no need of repeating a dimension, directly or indirectly. For example, the closing dimension has to be avoided if the total length is given.
6. For more details about dimensioning, refer to Chapter 4.

19.8 SUGGESTED DRAFTING PROCEDURE

To develop speed and accuracy in drawing, it is better to follow a certain order of drafting. All the instruments required for drawing should be placed at their proper locations in order to save time. The steps to be followed in making orthographic views are suggested below:

1. Decide the directions of the principal view (front view) and the combination of views such that it will best describe the object. Prepare freehand sketches of the required views and mark the overall dimensions on these views.
2. Considering the number of views to be drawn, with their overall dimensions and the size of the drawing sheet being given, select a suitable scale. But in industrial practice, the size of the drawing sheet is to be selected according to the number of views, overall dimensions and the scale of the drawing. The scale should be selected without spoiling the clarity of the drawing.
3. Draw the border line and outline of the title block. Leave sufficient space in between the views and the border line of the sheet. Care must be taken to provide necessary space for printing dimensions, notes, etc. [see Figure 19.13(b)]. As far as possible, provide equal space between the views for a better appearance.
4. Mark centre lines at appropriate places [see Figure 19.13(c)].
5. As far as possible, draw the details simultaneously in all the views. The following order of priority may be preferred as:
   1. Circles and arcs
   2. Straight lines which form the major shape of the object
   3. Straight lines, curves for the minor details like fillets, rounds, etc.
6. Draw all the details, except the hidden lines in all the views.
7. Erase all the unnecessary lines, construction lines, etc. Finish the drawing by thickening the appropriate lines.
8. Draw the hidden lines [see Figure 19.13(f)].
9. Enter all the dimensional values, distributing them appropriately in all the views.
10. Draw section lines, if any.
11. Name the views if necessary. Also enter other data necessary for the completion of the drawing. Check the drawing carefully and see whether there is any missing dimension, details, etc.
12. Print the title block details.

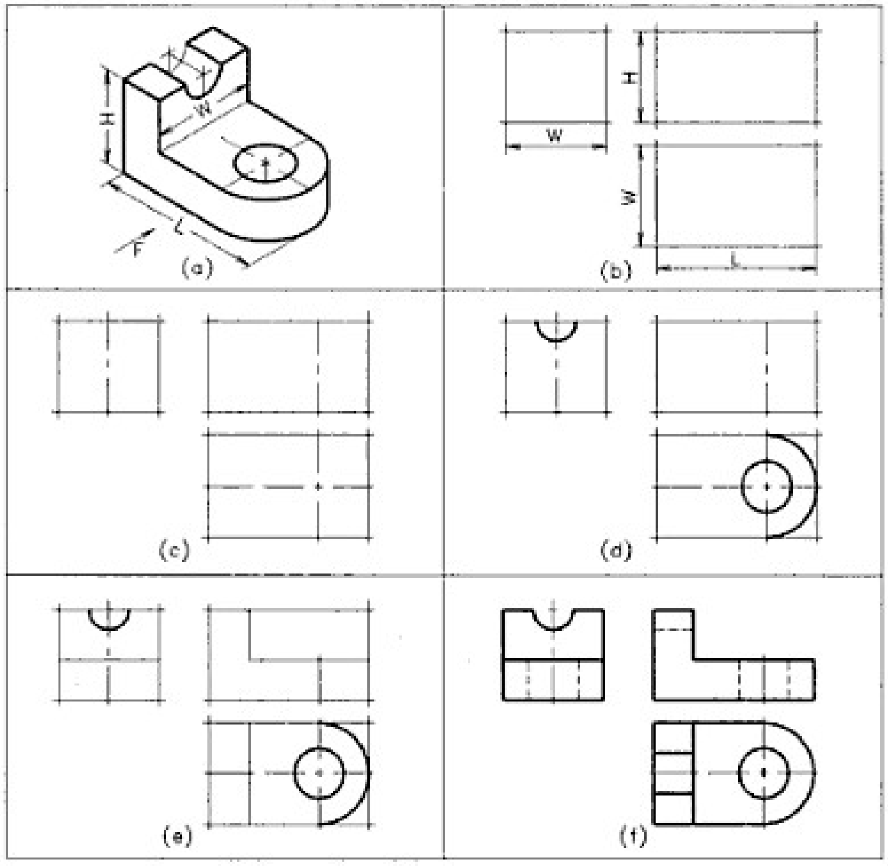


Figure 19.13 Drawing procedure for orthographic views.

Notes

1. Unless and otherwise specified, follow first angle projection method.
2. Symbol of the projection should be shown in the title block of the drawing sheet.
3. If third angle projection method is used along with first angle projection as a special requirement, it should be shown below the drawing by symbol or in writing.
4. Students are advised to name the orthographic views below them, until they develop the capacity to identify the views.
5. Projection lines and construction lines are not shown in orthographic views of objects.
6. Every circle should has two centre lines intersecting at the centre.
7. Axis of symmetry as well as axis of cylindrical holes, etc. should be drawn with the centre lines.
8. Choose a larger scale always for better clarity of views.
9. The hidden lines in a drawing should be minimised by orienting the object properly. Unimportant hidden details may be avoided, especially when the drawing is a complicated one.
10. While orienting an object for orthographic projection, the most important vertical face should be selected for the front view.
11. Front view is assumed to be the primary view for orthographic projection and all the remaining views are oriented in relation to the front view.
12. All the views should be drawn in the correct location with respect to front view as if there are projection lines. Shifted position of a view is assumed to be a spelling or grammar mistake, which will lead to wrong meanings in the graphic language.

19.9 FIRST ANGLE PROJECT OF OBJECTS HAVING PLANE SURFACES

Objects having plane surfaces alone may have the surface parallel to, inclined to or oblique to the reference planes. Figure 19.14 shows simple examples to these type of surfaces.

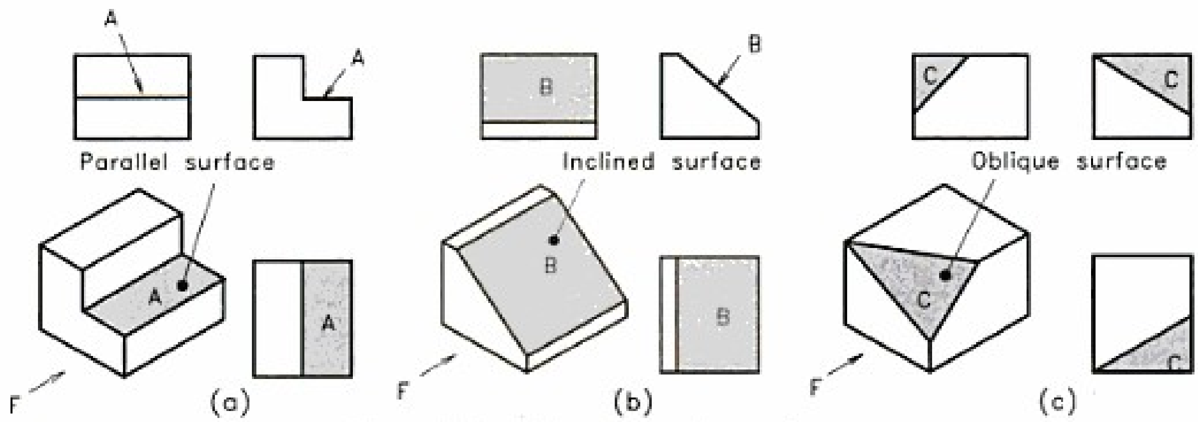


Figure 19.14 Objects having plane surfaces.

Parallel Surface

If a surface of an object is parallel to one of the reference planes of projection, the projection of the surface on that plane to which it is parallel will have its true size and shape. The projection of a surface which is normal to the plane of projection, is represented by a straight line.

Inclined Surface

If a surface of an object is perpendicular to one of the principal planes and inclined to the other two principal planes, the projections of the surface to which it is perpendicular will be a straight line, inclined to the other two reference lines. The projections of the surface on the reference planes to which it is inclined will appear foreshortened.

Oblique Surface

If a surface of an object is inclined to the three reference planes, the surface can be called as an *oblique surface*. If a surface is an oblique one, its projection will show areas on the three reference planes can be represented only by areas which will not give its true size and shape.

The following examples illustrate how orthographic views are drawn from pictorial views of objects having the above three types of plane surfaces.

Example 19.1

An isometric view of a parallel key is shown in Figure 19.15(a). Draw its front, top and left side views. The direction of the arrow, *F* shows the front side of the key.

Refer Figure 19.15(b). Follow the procedure explained in Section 1.8 to get the views.

Example 19.2

Draw the front, top and right side views of the angle-bracket shown in Figure 19.16(a).

Refer Figure 19.16(b). Follow the procedure explained in Section 19.8.

Example 19.3

Figure 19.17(a) shows an isometric view of a rectangular block having an oblique surface. Draw the front view looking in the direction of *F*. Add the top and the right side views.

Refer Figure 19.17(b). Follow the procedure explained in Section 19.8.

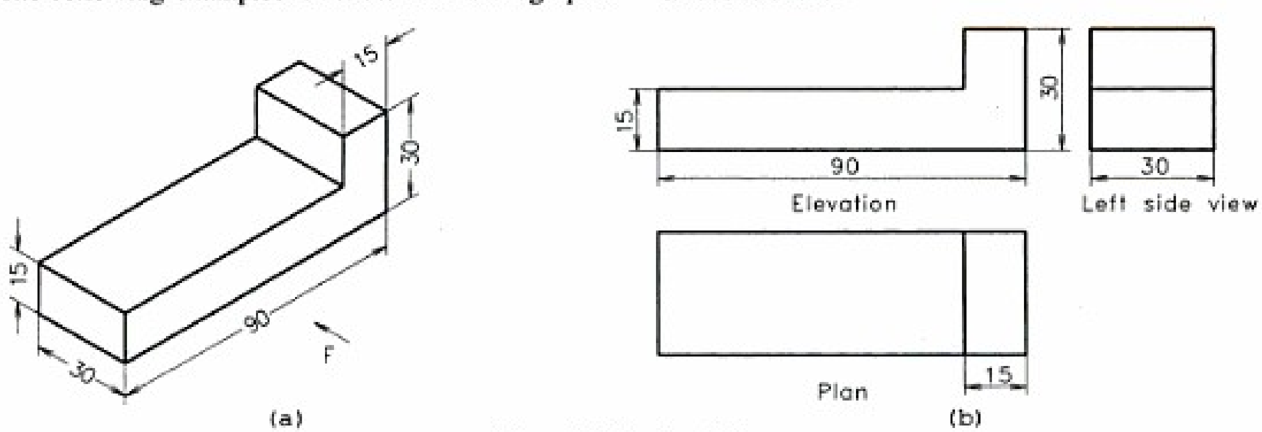


Figure 19.15 Parallel key.

weak. The radius of a fillet depends upon the thickness of the metal and other design requirements.

A rounded external corner on a casting is called *round*. External corners or angles are rounded for the appearance and comfort of persons who handle the casting.

Fillets and rounds actually prevent intersecting surfaces as they eliminate abrupt change in direction. This leads to certain problems in orthographic projections and the view becomes confusing (see Figure 19.19). To avoid this, lines are projected from approximate intersections.

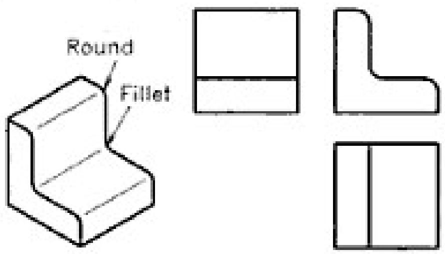


Figure 19.19 Fillets and rounds.

The radius of a fillet or round may sometimes be given on the view itself or as a general instruction. If the radius is not specified but the shape is shown in the given view, the radius may be assumed as 3 to 6 mm depending on the size of the object.

Example 19.4

Draw the three principal views of a cylindrical block shown in Figure 19.20(a).

Refer Figure 19.20(b).

Follow the procedure explained in Section 19.8.

Example 19.5

Isometric view of a shaft support is shown in Figure 19.21(a). Draw the front view, looking in the direction of the arrow *F*. Also draw top and the side views. Use a suitable scale.

Refer Figure 19.21(b).

Follow the procedure explained in Section 19.8.

Example 19.6

Isometric view of an object is shown in Figure 19.22(a). Draw the front view, looking in the direction of the arrow *F*. Also draw top and the side views. Use a suitable scale.

Refer Figure 19.22(b).

Follow the procedure explained in Section 19.8.

19.11 THIRD ANGLE PROJECTION OF OBJECTS

In the third angle system of projection, the object is assumed to be placed in the third quadrant and the views are obtained in the same side of viewing as explained in the Section 19.3. Except the change of position of views, there is no practical difference between first angle projection and third angle projection. In third angle projection, the top view is obtained on the top side of front view, the right side view is obtained on the right side of front view, and so on. This difference may be noted in the following example.

Example 19.7

Draw the front, top and right side views of an adjustable rod support shown in Figure 19.23(a). Use third angle projection method.

Refer Figure 19.23(b).

Draw the views as shown in figure, following the procedure explained above. Add an extra note "THIRD ANGLE PROJECTION" below the views.

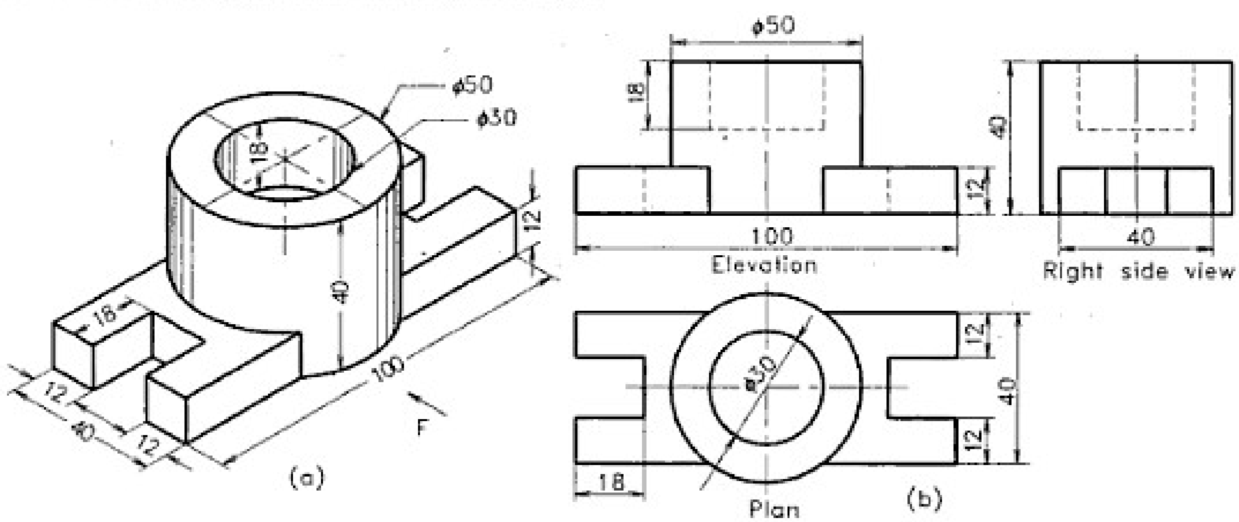


Figure 19.20 Cylindrical block.

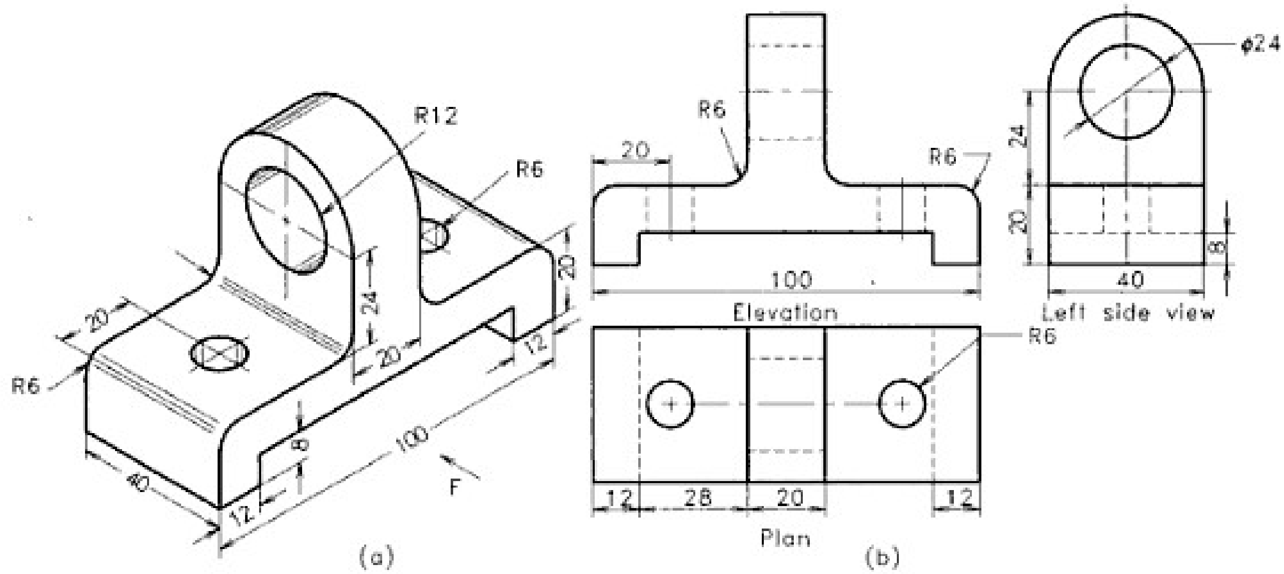


Figure 19.21 Shaft support.

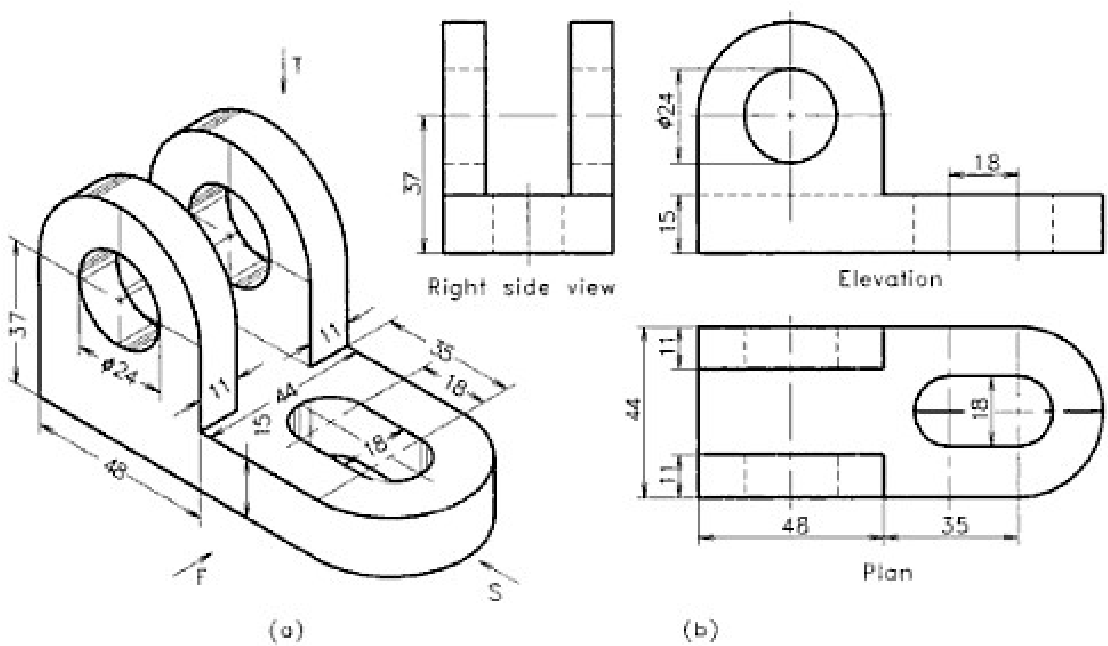


Figure 19.22 Bearing block.

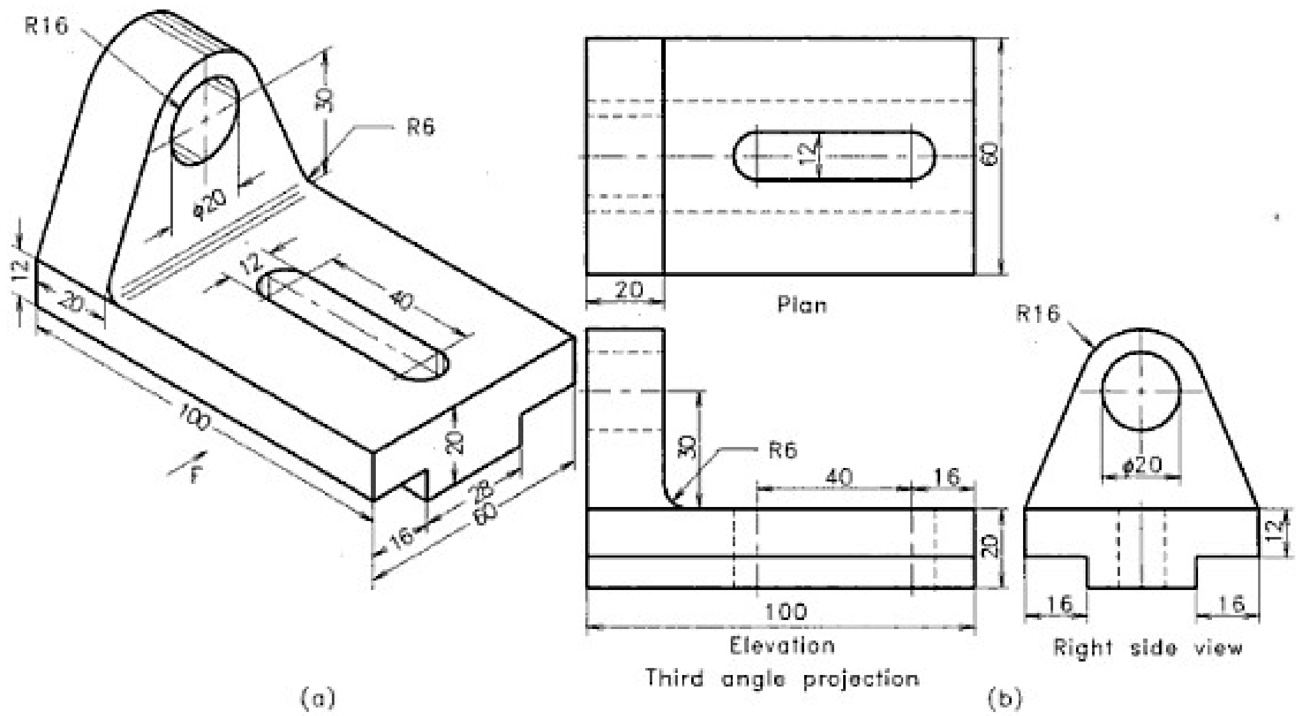


Figure 3.23 Adjustable rod support.

EXERCISES

Draw the orthographic views of the engineering objects as per the instructions given along with the Figures 19.24 to 19.40. View the objects in the direction of arrow marked with F. Name the view and dimension them as per BIS.

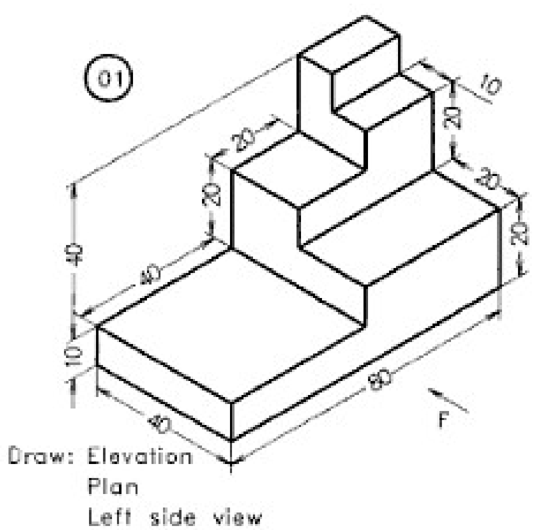


Figure 19.24 A block.

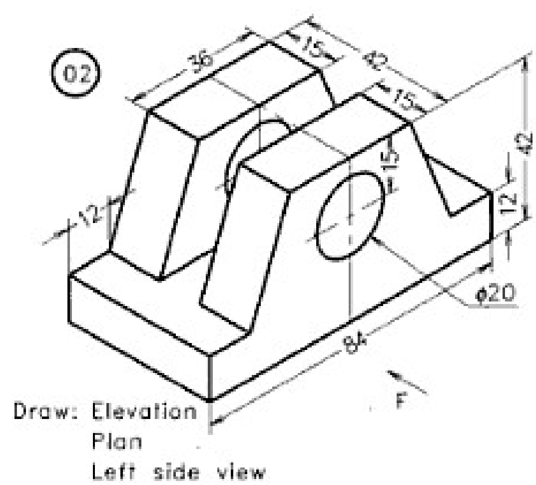
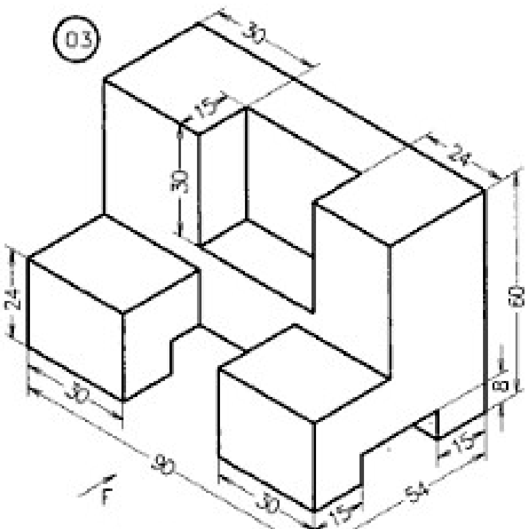
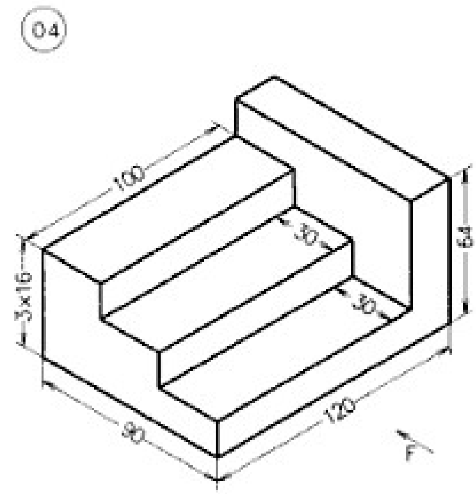


Figure 19.25 A block.



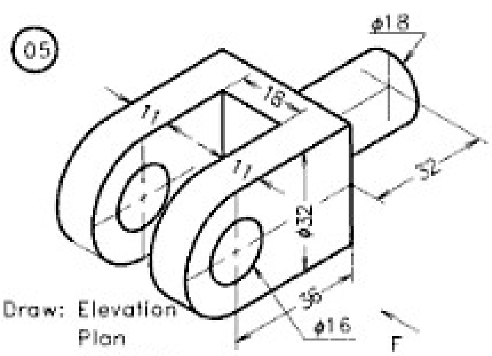
Draw: Elevation
Plan
Left side view

Figure 19.26 A block.



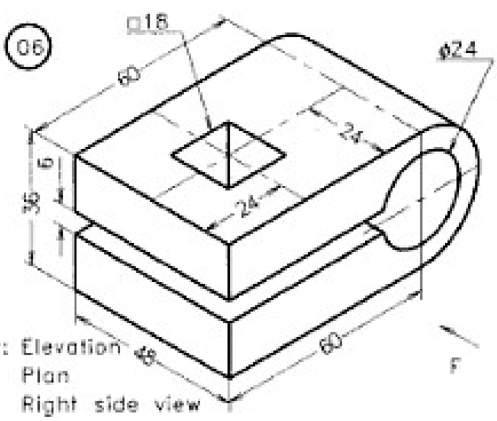
Draw: Elevation
Plan
Left side view

Figure 19.27 A stepped block.



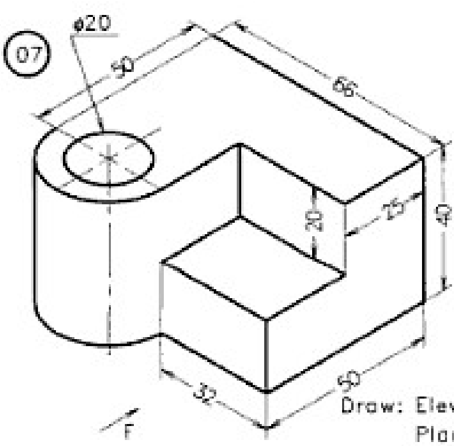
Draw: Elevation
Plan
Right side view

Figure 19.28 Fork end.



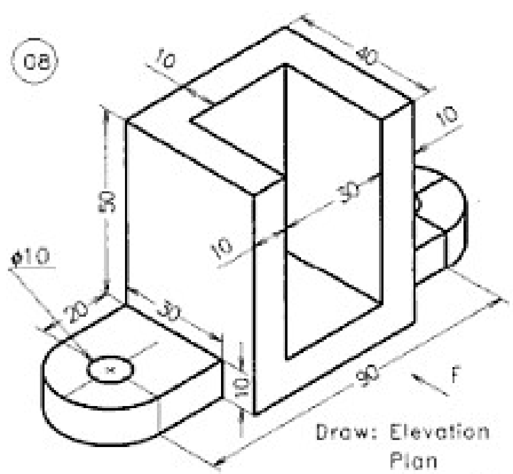
Draw: Elevation
Plan
Right side view

Figure 19.29 Fork end.



Draw: Elevation
Plan
Right side view

Figure 19.30 A cast iron block.



Draw: Elevation
Plan
Right side view

Figure 19.31 A stopper.

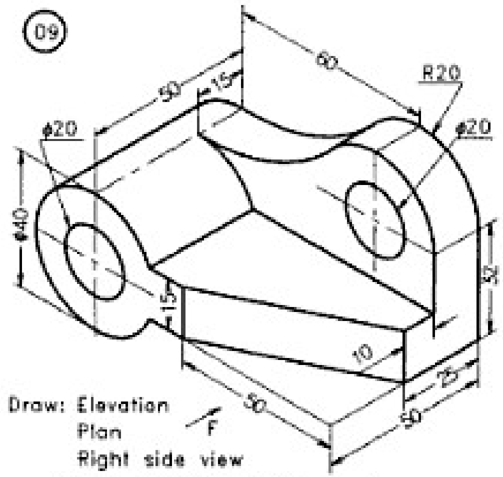


Figure 19.32 A shaft support.

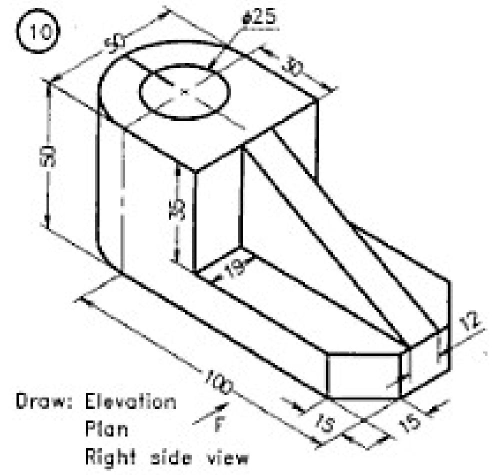


Figure 19.33 A block.

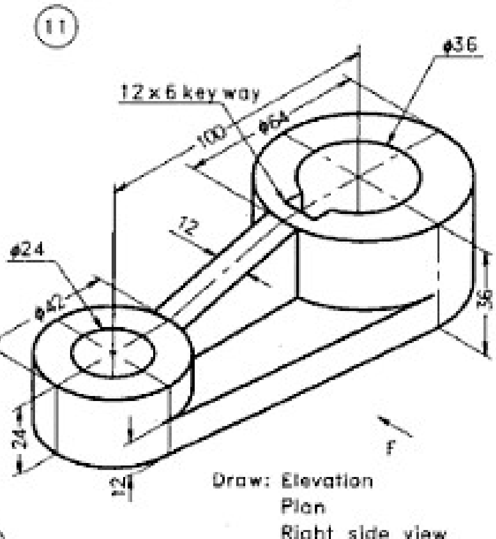


Figure 19.34 A lever.

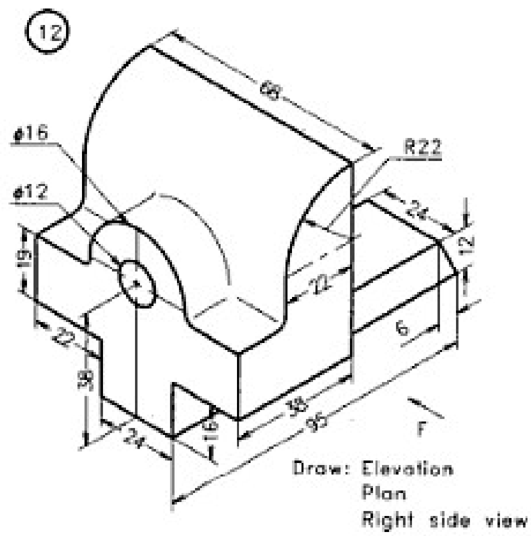


Figure 19.35 Jaw of a vice.

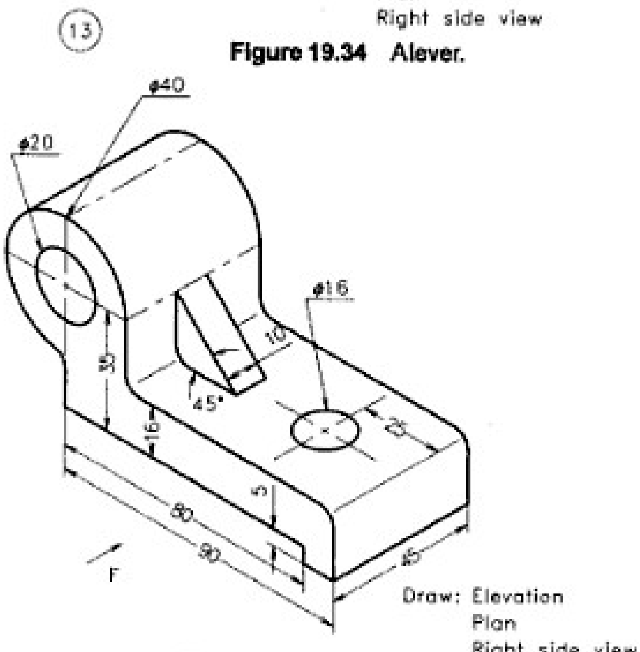


Figure 19.36 A bearing.

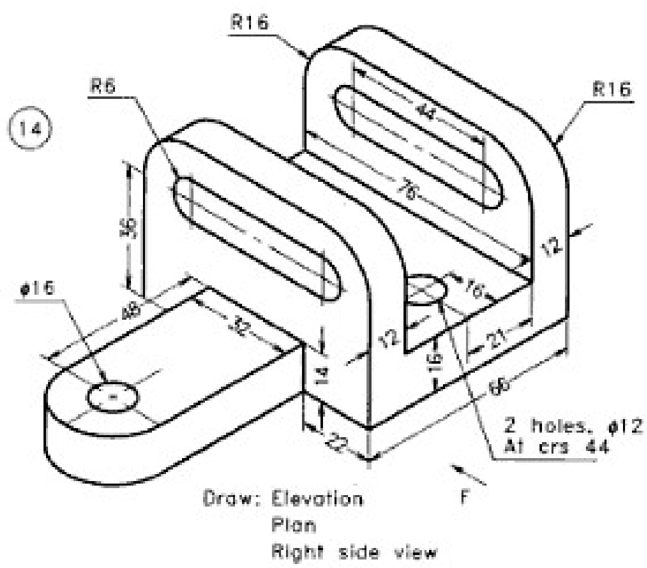


Figure 19.37 A bracket.